



Answered Promptly, Flagged All the Same: A Null-Result Test of Prompt-Response Speed Against Later Verification Flags on a Shared Streaming Account

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Abstract. A prompt asking a household to confirm that a shared subscription-video account is “still at your home” has become routine across the sector, and inside the platforms that deploy it the prompt is often treated informally as a light-touch trust signal: a household that answers quickly is a household with less to explain. We test that assumption directly. Working with a panel of 316 households sharing a subscription-video account across two or more physical addresses — share houses, holiday properties, and grandparents minding children among them — we classified each household’s response to its most recent home-confirmation prompt into a latency tercile (prompt, delayed, ignored) and followed the account for eleven months for any subsequent verification flag. Flag incidence was statistically indistinguishable across terciles (18.1–19.5%), as was time to first flag (median approximately 150 days in every group). An ablation excluding households with a flag already on record before the study began left the pattern unchanged. We report the null plainly: on this panel, how quickly a household answers a home-confirmation prompt predicts nothing about its subsequent flag history, and a verification system that reads prompt compliance as a durable trust signal may be crediting a household for very little.

Introduction

Subscription-video platforms increasingly gate continued access behind a periodic interstitial asking a household to confirm that the device in use is “still at your home” — a light prompt that appears on a smart-TV interface or a companion app, typically resolved with a single tap. The prompt sits at the intersection of two institutional interests: a platform’s interest in constraining how far a single account travels, and a household’s interest in getting back to whatever it was about to watch. How a household responds to the prompt is, in practice, one of the few interaction traces a verification system has to work with, and it is tempting — inside product teams, and increasingly inside research building on their published methodology — to read responsiveness as a proxy for account legitimacy: a household that answers quickly is a household with nothing to explain.

This paper treats that reading as an empirical question rather than a design assumption. If prompt-response speed functions as the light-touch trust signal it is often taken to be, households that respond fastest should accumulate fewer follow-up verification flags, and should accumulate them later, than households that delay or ignore the prompt outright. If the prompt instead measures something closer to a household’s general attentiveness to app notifications — a trait with no particular bearing on where an account is actually used — no such pattern should appear.

We report an eleven-month panel study of 316 multi-address households sharing a subscription-video account, linking each household’s response latency to its most recent home-confirmation prompt against its subsequent verification-flag history.

Three contributions follow, each stated with the caution the design supports.

- We assemble a household-level panel directly linking home-confirmation prompt-response latency to later account-verification outcomes, a linkage we are not aware of having been reported before.
- We find no relationship between the two, across three complementary measures — flag incidence, time to first flag, and a robustness check excluding households with a pre-existing flag — and report this null without recasting it as a weak positive.
- We suggest, carefully and only as far as this single panel supports, that a verification system built around prompt compliance may be formalising a signal with less content than its ubiquity implies.

Related work

Security and privacy scholarship has long questioned whether a user’s response to an interstitial prompt carries the informational weight its design assumes. Habituation research finds that repeated, static security warnings lose their behavioural effect within a handful of exposures, with redesigned, polymorphic warnings partially restoring attention [1]; earlier mental-model work made the related point that a warning’s meaning to its designer and its meaning to the user who dismisses it routinely diverge [2], and design guidance for browser security and privacy alerts has, on this basis, pushed toward differentiating a prompt’s visual weight by the severity of what it gates rather than presenting every interstitial identically [3]. Consent-notice research reports a parallel pattern: GDPR consent choices are driven far more by interface defaults — button placement, pre-selected options — than by users engaging with a notice’s actual content [4], automated tooling built to detect and classify consent notices at scale finds the same handful of interaction patterns recurring across otherwise unrelated sites [5], automated audits find consent-revocation paths frequently harder to complete than the original consent well after the regulatory deadline for parity [6], and studies of cookie-banner design show response speed tracking the banner’s layout more than any underlying preference [7]. Non-expert users, in their own words, describe everyday security prompts more often as an obstacle to clear than as information to weigh [8], and attitudinal

measures of trust in an automated system are known to diverge from behavioural reliance on that same system [9].

The “is this your home” prompt sits on infrastructure with its own documented unreliability: IP-geolocation databases, the usual signal behind an automated household-location check, drift and misclassify over time [10], with measurable downstream consequences where they are relied on [11], and automatic content recognition on smart televisions already extracts household viewing signals independent of any prompt [12]. Multi-user smart-home access-control research faces the structurally similar problem of attributing a shared device to the right household member [13], and a broader economics-of-privacy literature situates a household’s choices about how narrowly to hold access to a shared digital good within incentive structures rather than awareness [14] — consistent with fieldwork on intra-household management of jointly held resources, which documents ongoing within-household negotiation over a shared good rather than a single unilateral decision [15].

Method

Setting. The panel comprises 316 households, recruited through the Adaptive Metrics Lab’s standing household-research registry, each sharing a single subscription-video account across two or more physical addresses at enrolment — share houses with a common account holder, holiday properties used seasonally, and grandparents’ households minding children on a recurring basis were the three commonest configurations. Every household had encountered at least one “is this your home” prompt on the shared account in the three months before enrolment.

Response-latency coding. For each household h , we recorded the interval between the platform serving its most recent home-confirmation prompt and the household’s response:

$$T_h = t_{\text{response}}(h) - t_{\text{prompt}}(h),$$

right-censored at 14 days for households that never responded. Households were sorted into terciles on T_h : **Prompt** ($T_h < 24$ hours, $n = 104$), **Delayed** (24 hours – 7 days, $n = 118$), and **Ignored** ($T_h > 7$ days or no response recorded, $n = 94$).

Follow-up. Each household's account was tracked for eleven months following tercile assignment for any subsequent verification flag — an additional identity or location check the platform imposes on an account, drawn from the household's own account-activity log with the household's consent. A household could contribute at most one flag event to the primary analysis (time to first flag); flag incidence is a binary indicator of whether any flag occurred in the window.

Flag rate. For tercile group g ,

$$\text{FlagRate}_g = \frac{N_{\text{flag}}(g)}{N_{\text{total}}(g)},$$

compared across groups with a chi-square test of independence; time to first flag was compared with a log-rank test, with non-flagged households right-censored at eleven months.

Ablation. Forty-one households (13.0% of the panel) had at least one verification flag already on record before enrolment. The primary analysis retains them; an ablation repeats both comparisons on the remaining 275 households to check whether a pre-existing flag history, rather than prompt-response speed itself, was driving any group difference.

Results

Flag incidence holds flat across response speed. Figure 1 compares flag incidence across the three terciles: 18.3% of Prompt households (19/104), 19.5% of Delayed households (23/118), and 18.1% of Ignored households (17/94) recorded at least one verification flag within the eleven-month window ($\chi^2(2) = 0.09$, $p = .96$). The fastest-responding tercile did not accumulate measurably fewer flags than the tercile that never responded at all.

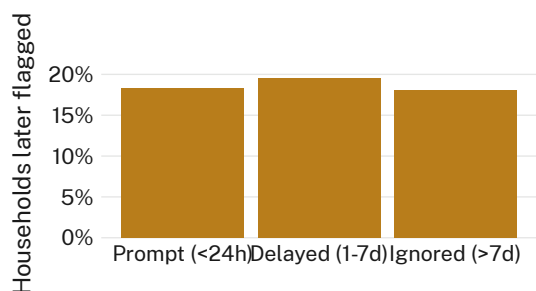


Figure 1: Flag incidence held between 18.1% and 19.5% across all three prompt-response terciles (n.s.).

Time to first flag holds flat too. Figure 2 plots the distribution of days to first flag by tercile among the 59 households that flagged. Median time to first flag was 152 days for Prompt households, 149 days for Delayed households, and 145 days for Ignored households, with overlapping interquartile ranges throughout; a log-rank test across all three groups, including right-censored non-flagged households, found no difference in the underlying hazard ($\chi^2(2) = 0.41$, $p = .81$).

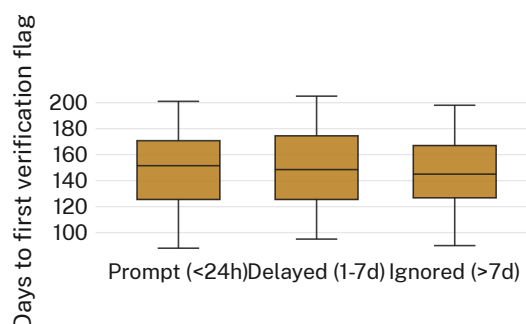


Figure 2: Days to first verification flag show the same spread across all three terciles, whether a household answered within the hour or never at all.

The pre-existing-flag ablation changes nothing further. Excluding the 41 households with a flag already on record before enrolment shifts overall flag incidence from 18.7% to 18.4% and leaves the tercile-level pattern in Figure 1 within half a percentage point at every tercile ($\chi^2(2) = 0.11$, $p = .95$ on the reduced sample); we retain the full panel for the primary analysis, since a household's flag history before this study began is not a quantity the prompt itself was designed to predict.

One process detail held constant across every flag event the panel recorded: in all 59 cases, the platform's own review of a raised flag completed within its published 72-hour service window, with no case exceeding it, regardless of which tercile the flagged household belonged to. Review speed, in other words, tracked the platform's own internal service target and nothing this study measured about the household on the other end of the flag.

Discussion

Read together, the three results locate the home-confirmation prompt's informational content precisely: it has none, for the outcome a verification system is presumably built to antic-

ipate. A household's willingness to tap through a prompt within the hour predicts neither whether it is later flagged nor how long that flag takes to arrive, and excluding households with a flag history the prompt could not have seen changes nothing about that picture. Prior work at this University found a structurally similar null when auditing a different verification signal against ground truth: a suburban pool's smoke-day flag showed no relationship to same-day attendance or symptom reports either [16]. A companion paper offers the sharper contrast: requiring a childcare room to actively re-confirm a settling-in flag, rather than letting it carry over by default, did change how long that flag outlived its own justification [17]. The difference between the two designs is instructive. Re-confirmation there was a structural requirement built into the flag's own lifecycle; here it is a single voluntary tap a household can complete without engaging with anything the prompt is meant to establish. It may be the mechanism's design, not a household's responsiveness to it, that carries whatever signal a verification system is looking for.

Limitations

The panel draws on one subscription-video account category and one platform's prompt design; though the null held identically across three household configurations (share house, holiday property, grandparent-minded), we cannot rule out that a differently worded or differently timed prompt would correlate with flag history where this one does not. Flag events were drawn from household-reported account-activity logs rather than the platform's own internal audit trail, a proxy we consider adequate given the flag event itself is a visible, dated notification no household in the panel reported missing. The eleven-month window captures a single flag cycle for most households; a panel followed across a full renewal cycle might yet find a relationship this window is too short to surface, though the flatness already visible at day 150 gives that possibility little obvious room to hide in.

Conclusion

Across an eleven-month panel of 316 multi-address households, how quickly a household answered a shared subscription-video account's home-confirmation prompt predicted nothing

about whether or when that account was later flagged for further verification. Flag incidence held flat within two percentage points across every response-speed tercile, time to first flag held flat within a week of its own median, and excluding households with a flag history the prompt could not have seen changed neither pattern. The distinction matters for any verification system crediting prompt compliance as evidence of household legitimacy, when the more defensible reading is that the prompt measures little beyond a household's willingness to tap a button. Future work should test whether the prompt's wording, rather than its mere presence, is where any signal would have to live.

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